

Total No. of Questions : 8]

SEAT No. :

P826

[5870] - 1149

[Total No. of Pages : 2

T. E. (Information Technology)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester - II) (314452)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1, or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume the suitable data, if necessary.

- Q1)** a) Explain Big data Ecosystem with suitable diagram. [7]
b) Explain anatomy of File read and write in HDFS. [7]
c) Write and explain any two Hadoop shell commands. [4]

OR

- Q2)** a) Explain Map Reduce with proper diagram for word count example. [7]
b) Explain Google file system. [7]
c) Explain ETL processing. [4]

- Q3)** a) Explain different steps in Data Analytics Project Life cycle [7]
b) Draw and explain Architecture of HIVE. [7]
c) Explain different data transformation techniques. [3]

OR

- Q4)** a) Explain different kinds of Big Data Analysis. [7]
b) How data can be ingested in python. Write syntax in python for the same. [7]
c) Explain role of visualization in big data analytics. [3]

- Q5)** a) Explain different techniques of Big Data visualization. [7]
b) Explain challenges in Big data visualization. [7]
c) Write two data visualization functions from matplotlib. [3]

OR

P.T.O.

- Q6)** a) Explain different tools for data visualization. [7]
b) Explain scatter plot, histogram and heat map with example. [7]
c) Write two data visualization functions from seaborn. [3]

- Q7)** a) How Social Media analytics helps in value creation? Explain with suitable examples. [7]
b) Explain in brief data analytics life cycle. [7]
c) Explain big data value terminology. [4]

OR

- Q8)** a) What is text mining? Draw and explain text mining architecture and its use. [7]
b) Explain Big data analytics in research. [7]
c) Explain big data impact on organizations. [4]

Total No. of Questions : 8

SEAT No. :

PA-1508

[Total No. of Pages : 2

[5926]-128

T.E. (Information Technology)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester-II) (314452)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) i) Explain Google File system and its advantages. [5]
ii) Explain ETL in Big data. [5]
b) Explain Hadoop distributed file system. [8]

OR

- Q2)** a) i) Write 5 Hadoop Shell commands. [5]
ii) Explain Role Job tracker and Task Tracker in Hadoop Architecture. [5]
b) Explain Map Reduce with proper diagram for word count example. [8]

- Q3)** a) Explain different types of Big Data Analysis techniques. [8]
b) i) Explain Different Data Transformation techniques. [3]
ii) What is dataset? Explain with python syntax of 2 different types of dataset used in Big data. [6]

OR

- Q4)** a) i) Explain Mean, Mode and variance and standard deviation with suitable example. [8]
ii) Explain Data Standardization. [3]
b) Draw and explain Architecture of HIVE. [6]

P.T.O.

- Q5)** a) i) How data visualization help Big data Analytics. [4]
ii) List the conventional Data visualization tools. Explain any Two. [6]
b) Explain data visualization with the help of example? What are the advantages of data visualization? [8]

OR

- Q6)** a) Explain any 4 Types of data visualization with example. [8]
b) i) Explain different data visualization tools. [6]
ii) Explain Data Visualization with Tableau. [4]

- Q7)** a) Explain Text mining with example. [8]
b) Explain Big Data Analytics Challenges in brief. [9]

OR

- Q8)** a) Explain four Big Data use cases. [8]
b) Explain types of Mobile Analytics. [9]



Total No. of Questions : 8]

SEAT No. :

P490

[Total No. of Pages : 2

[6003]-711

T.E. (I.T.)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester-II) (314452)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

Q1) a) Explain Google file system and its advantages. [10]

b) Explain Hadoop distributed file system [8]

OR

Q2) a) Why map reduce is required in Hadoop? Explain the stages involved in map reduce task with a suitable example? [9]

b) Describe the various types of NoSQL Databases with example and also compare them. [9]

Q3) a) Explain Mean, Mode and variance and standard deviation with suitable example. [9]

b) Draw and explain Architecture of HIVE [8]

OR

Q4) a) Explain Min-max scaling. For the following dataset carry out min-max Scaling, X=24, 28, 53, 30, 40, 18, 15, 21 [9]

b) What is data Wrangling? Why do you need it? explain data Wrangling methods? [8]

Q5) a) Explain any 4 Types of data visualization with example. [9]

b) Explain different data visualization tools. [9]

OR

P.T.O.

- Q6) a)** Explain data visualization with the help of example? What are the advantages of data visualization? [9]
- b)** Explain Data Visualization with Tableau. [9]

- Q7) a)** Explain Big Data Analytics Challenges in brief. [9]
- b)** Explain types of Mobile Analytics. [8]

OR

- Q8) a)** What is Porter's value chain analysis? Explain Porter's value chain analysis. [9]
- b)** What is social media analytics? Explain the process of social media data analytics. [8]



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SEAT No. :

P-7626

[Total No. of Pages : 2

[6180]-146

T.E. (Information Technology)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - II) (314452)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*

Q1) a) Explain the process of reading and writing a file in HDFS with neat diagram. [8]

b) List and explain any four Hadoop shell commands with syntax. [4]

c) Differentiate between SQL and NoSQL databases with example. What is the need to develop big data applications using NoSQL databases? [6]

OR

Q2) a) What is the need of map reduce in Big Data? Explain the stages involved in map reduce task with a suitable example? [9]

b) Explain Hadoop ecosystem in detail. [9]

Q3) a) What is data wrangling? Why do you need it? Explain data wrangling methods? [9]

b) What is categorical variable? Why do you need categorical variable encoding? With an example, explain one-hot encoding. [9]

OR

Q4) a) Draw and explain Architecture of HIVE. [8]

b) How missing values and categorical variables are preprocesses before building model? Explain with suitable example. [4]

c) Explain z-score normalization. For the following dataset carry out z-score normalization (standardization), X = 23, 29, 52, 31, 45, 19, 18, 27. [6]

P.T.O.

- Q5)** a) What is Data Visualization? What are the major challenges in big data visualization and how to overcome these challenges? [6]
- b) Explain various techniques for visual data representation. [6]
- c) Explain the following data visualization techniques. [5]
- i) Candela
- ii) D3.js

OR

- Q6)** a) Explain data visualization with respect to 1-D, 2-D, 3-D data. [6]
- b) Explain various analytical techniques used in big data visualization. [6]
- c) Draw histogram with a suitable example and explain its usage. [5]
- Q7)** a) What is Porters valuation creation model? Explain porter's value chain analysis. [9]
- b) What is social media analytics? Explain the process of social media data analytics. [8]

OR

- Q8)** a) What is text mining? Draw and explain text mining architecture and its use. [8]
- b) Explain primary activities of Michal Porters value chain. [5]
- c) How mobile analytics is different from social media analytics? [4]



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SEAT No. :

PB3866

[Total No. of Pages : 2

[6262] 130

T.E. (Information Technology)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - II) (314452)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Figures to the right indicate full marks.

- Q1) a)** Explain all the steps for writing a file in HDFS with neat diagram. **[8]**
- b)** Describe the various types of NoSQL Databases with example and also compare them. **[10]**

OR

- Q2) a)** Why map reduce is required in Hadoop? Explain the stages involved in map reduce task with a suitable example? **[9]**
- b)** What is Hadoop Distributed system? What is the advantage of heart bit message in Hadoop. **[9]**
- Q3) a)** Compare HBASE and HIVE with suitable parameters. **[8]**
- b)** How missing values are filled in Pandas Data Frame with zeros? Assume suitable data. **[3]**
- c)** Explain Min-max scaling. For the following dataset carry out min-max Scaling, X = 24,28,53,30,40,18,15,21. **[6]**

OR

- Q4) a)** What is categorical variable? Why do you need categorical variable encoding? With an example, explain one-hot encoding? **[8]**
- b)** What is data wrangling? Why do you need it? Explain data wrangling methods? **[9]**

P.T.O.

- Q5)** a) How Data Visualization is important in Big Data? Explain challenges to big data visualization? [6]
- b) Explain various techniques for visual data representation. [6]
- c) Explain the following data visualization techniques. [6]
- i) Google Chart API
- ii) D3.js

OR

- Q6)** a) Explain data visualization with respect to 1-D, 2-D, 3-D data? [6]
- b) Explain various analytical techniques and tools used in data visualization. [6]
- c) Draw boxplot with a suitable example and explain its usage. [6]
- Q7)** a) What is text mining? Draw and explain text mining architecture and its use. [8]
- b) Explain primary activities of Michael Porter's value chain. [5]
- c) How mobile analytic is different from social media analytic? [4]

OR

- Q8)** a) What is Porter's value chain creation model? Explain Porter's value chain analysis. [9]
- b) What is social media analytic? Explain the process of social media data analytic. [8]

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Total No. of Questions : 8]

SEAT No. :

PC1809

[6353]-128

[Total No. of Pages : 2

T.E. (Information Technology)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - II) (314452)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Figures to the right indicate full marks.

Q1) a) Explain the working of Google File System with a neat diagram with relevant to Big Data. [9]

b) Write Short note on YARN and Map - Reduce technique. Is there any difference between both. [8]

OR

Q2) a) Explain Hadoop Frame work components with a neat Diagram, which component is used in Big data. [9]

b) Write the 4 types of NoSQL databases. How they are used in Big Data. [8]

Q3) a) Explain 4 types of big data analytics along with the analytical purpose in each phase. [8]

b) Explain different steps in Data Analytics project life cycle, how it is different than normal project life cycle. [9]

OR

Q4) a) Explain the tasks in Data integration and Data Transformation phase with suitable example. [8]

b) Explain the three ways of Data Normalization used in data analytics. [9]

P.T.O.

Q5) a) Explain different Data Visualization Techniques. Why visualization is important in Big Data. **[8]**

b) Explain various types of charts used in data visualization along with its analytical purpose. **[10]**

OR

Q6) a) Explain how data Visualization is beneficial for Business Analytics and Intelligence. **[8]**

b) Explain Features of Tableau DV tool and state phases of Tableau workflow. **[10]**

Q7) a) What is Social Media Data Analytics and state the advantages of SMA. **[9]**

b) Explain roles and responsibilities of big data analysts and data scientists. **[9]**

OR

Q8) a) What is Mobile analytics? Explain its importance with suitable example. **[9]**

b) Explain the data analysis life cycle in big data. **[9]**



Total No. of Questions : 8]

SEAT No. :

PD4330

[Total No. of Pages : 2

[6403]-128

T.E. (Information Technology)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - VI) (314452)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*

- Q1)** a) Explain GFS as solution for Big data Processing Challenges. [6]
- b) Write short note on HIVE and HBASE. [8]
- c) Explain any 3 Hadoop shell commands. [3]

OR

- Q2)** a) Explain HDFS with a neat diagram. [6]
- b) Write the 4 types of NoSQL databases. [8]
- c) State and explain CAP Theorem. [3]

- Q3)** a) Explain the 4 types of big data analytics along with the analytical purpose. [8]
- b) Explain the different modes of data transformation in big data. [9]

OR

- Q4)** a) Explain the tasks in Data integration and Data Transformation phase. [8]
- b) Explain the three ways of Data Normalization used in data analytics. [9]

P.T.O.

- Q5)** a) Explain different Data Visualization Techniques. [8]
b) What are five purposes of using charts and state which type of charts used for each purpose? [10]

OR

- Q6)** a) Explain how data Visualization is beneficial for Business Analytics and Intelligence. [8]
b) Write note on any three of the following DV tools. [10]
i) Tableau
ii) Infogram
iii) Fusion Charts

- Q7)** a) Explain what is Social Media Data Analytics and state the advantages of SMA. [9]
b) Explain the data analysis life cycle in big data with example. [9]

OR

- Q8)** a) What is Mobile analytics? Explain its importance with suitable example. [9]
b) Explain Text mining architecture with suitable example. [9]

